

Amrita Raghunath

Machine Learning Researcher — Computer Vision, Multimodal Sequence Modeling, Sign Language & Accessibility AI

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Research Summary

Machine Learning Researcher with 6+ years of experience specializing in multimodal deep learning systems, with a core focus on AI accessibility, linguistic inclusion, and marginalized language technologies. Experienced in architecting end-to-end neural network architectures spanning real-time computer vision and natural language processing. Key contributions include engineering neural machine translation (NMT) pipelines tailored to low-resource languages in India, as well as developing and deploying a government-funded Indian Sign Language (ISL) real-time chatbot. Currently an independent researcher investigating brain-computer interfaces (BCIs), actively seeking PhD and advanced research opportunities in assistive neurotechnology and inclusive AI.

Education

Master of Technology (MTech), Artificial Intelligence

2019 – 2021

Amrita Vishwa Vidyapeetham

Kollam, India

- CGPA: 8.51/10 | Thesis Advisor: Prof. Geetha M | Focus: Object detection and classification, Image to text translation

Bachelor of Technology (BTech), Computer Science

2014 – 2018

Amrita Vishwa Vidyapeetham

Kollam, India

- CGPA: 8.53/10 | Project Advisor: Prof. Krishnaprasad TR | Focus: Peer-To-Peer Network
- Exchange Semester, University of Paderborn, Germany (2018) — Thesis Advisor: Prof. Holger Karl | Focus: Network Function Virtualization

Publications

- M. Geetha, N. Aloysius, D. A. Somasundaran, **A. Raghunath**, and P. Nedungadi, "Toward Real-Time Recognition of Continuous Indian Sign Language: A Multi-Modal Approach Using RGB and Pose," *IEEE Access*, vol. 13, pp. 60270–60283, 2025. doi: 10.1109/ACCESS.2025.3554618
 - Proposed **SignFlow**: domain-specific pre-training (3D CNN on isolated ISL videos; Transformer on MediaPipe pose/hand estimates) with dynamic downsampling (6 fps) and CTC loss
 - Achieved Word Error Rate (WER) of 19 on continuous ISL recognition, with a novel detection-rate metric for keyword identification

Research Experience

Independent Researcher

Mar 2024 – Present

Self-directed

Kerala, India

- Finalized and published the SignFlow continuous ISL recognition paper in *IEEE Access* (2025)
- Scoping a new research direction in brain-to-text implementations, inspired by Brain2Qwerty-style architectures, learning about ways to decode natural sentences from non-invasive neural modalities like magnetoencephalography (MEG) recordings.

Machine Learning Engineer

Jan 2022 – Mar 2024

Amrita Center for Research in Analytics, Technologies & Education (CREATE)

Kollam, India

- Developed a Conformer-based continuous ISL recognition model in PyTorch, improving on a prior CNN-Transformer baseline through hyperparameter optimization and refined evaluation metrics
- Designed and implemented a CNN-Transformer continuous sign recognition pipeline (isolated-to-continuous transfer pretraining), incorporating MediaPipe/OpenCV-based pose generation and data augmentation
- Built an FAQ-retrieval module for ISL query handling using clustering algorithms and a custom question-answer mapping algorithm, integrated into a government-funded ISL chatbot for public-service accessibility
- Constructed a benchmark static ISL dataset (alphabets, digits 0–9) with MediaPipe/OpenCV preprocessing; benchmarked VGG and ResNet classifiers in PyTorch

Research Associate

Mar 2021 – Nov 2021

Amrita Centre for Cyber Security Systems and Networks

Kollam, India

- Contributed to projects like development of malware detection systems and malware reverse engineering using ML.

Junior Research Fellow

Dec 2018 – Jul 2019

Amrita Centre for Cyber Security Systems and Networks

Kollam, India

- Developed an unsupervised neural machine translation system for Mizo language using semi-supervised training (BERT encoder, LSTM decoder) over monolingual Mizo/Nagamese corpora
- Automated monolingual corpus collection via Selenium web crawling; trained FastText, GloVe, and MUSE word embeddings

Selected Earlier Projects

- Semantic-Guided Scene Description for the Visually Impaired** (2021) — Image captioning model fusing visual-semantic information via Faster-RCNN, BERT, LSTM, and Transformer decoders
- Real-Time Face Detection for Crowd Surveillance** (2019) — YOLO-based face detection/recognition for CCTV using PixelLib/OpenCV

Teaching & Mentorship

- **Teaching Assistant**, Object-Oriented Programming (Python), BTech program, Amrita Vishwa Vidyapeetham (2020–2021), under Prof. Neetha J.
- **ISL Accessibility Workshop** (2023) — Workshop for the ISL community with CDAC, Trivandrum, demonstrating the ISL chatbot
- **Leadership**, Ayudh National Leadership Summit (2018) — Mentored 30+ participants

Technical Skills

- **Languages**: Python, C++, C, Java, JavaScript, Bash, MATLAB
- **ML/DL Frameworks**: PyTorch, TensorFlow, Keras, HuggingFace, Scikit-learn
- **CV & Sequence Modeling**: Computer Vision, NLP, Generative Models, Reinforcement Learning, MediaPipe, OpenCV
- **Data & Infrastructure**: NumPy, Pandas, PostgreSQL, MySQL, AWS, Docker, Git, Jupyter, Google Colab, Neptune.ai
- **Other**: LaTeX, ImgLab, Linux

Certifications & Languages

- **Certification**: Deep Learning Specialization, Coursera (May 2020)
- **Languages**: English (TOEFL 100/120), German (A2), Malayalam (Native), Sanskrit (Conversational)